Answer Key - Mathematics - Graduate Level Assessment 21 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** E
   Options: A) 0; B) 4; C) 3; D) 2; E) 9
   Note: Correct option: E - 9

2. **Answer:** B
   Options: A) f has all of its relative extrema on the line y = x^2.; B) f has all of its relative extrema on the line x = y.; C) f has all of its relative extrema on the line x = 2 y.; D) f has a relative maximum at (0, 0).; E) f has an absolute maximum at (2/3, 2/3).
   Note: Correct option: B - f has all of its relative extrema on the line x = y.

3. **Answer:** A
   Options: A) The sum of the residuals is always one.; B) If the correlation is 0, there will be a distinct pattern in the residual plot.; C) The larger the residuals, the better the linear model.; D) The residuals are always negative.; E) If the correlation is 1, there will be a distinct pattern in the residual plot.
   Note: Correct option: A - The sum of the residuals is always one.

4. **Answer:** A
   Options: A) 19.0; B) 17.0; C) 25.0; D) 23.0; E) 18.0
   Note: Correct option: A - 19.0

5. **Answer:** C
   Options: A) 20 over 28; B) 2 and 3 over 14; C) 1 and 3 over 7; D) 2 and 1 over 14; E) 3 and 2 over 7
   Note: Correct option: C - 1 and 3 over 7

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** From the graph of $xy = 1$, we can tell that the foci will be at the points (t, t) and $(-t, -t)$ for some positive real number t. [asy] unitsize(1 cm); real func(real x) { return(1/x); } pair P; pair[] F; P = (1/2,2); F[1] = (sqrt(2),sqrt(2)); F[2] = (-sqrt(2),-sqrt(2)); draw(graph(func,1/3,3),red); draw(graph(func,-3,-1/3),red); draw((-3,0)–(3,0)); draw((0,-3)–(0,3)); draw(F[1]–P–F[2]); dot("F_1", F[1], SE); dot("F_2", F[2], SW); dot("P", P, NE); [/asy] Thus, if $P = (x, y)$ is a point on the hyperbola, then one branch of the hyperbola is defined by

$$\sqrt{(x + t)^2 + (y + t)^2} - \sqrt{(x - t)^2 + (y - t)^2} = d$$

for some positive real number d. Then

$$\sqrt{(x + t)^2 + (y + t)^2} = \sqrt{(x - t)^2 + (y - t)^2} + d.$$

Squaring both sides, we get

$$(x + t)^2 + (y + t)^2 = (x - t)^2 + (y - t)^2 + 2d\sqrt{(x - t)^2 + (y - t)^2} + d^2.$$

This simplifies to

$$4tx + 4ty - d^2 = 2d\sqrt{(x - t)^2 + (y - t)^2}.$$

Squaring both sides, we get &16 t^2 x^2 + 16 t^2 y^2 + d^4 + 32 t^2 xy - 8 d^2 tx - 8 d^2 ty &= 4 d^2 x^2 - 8 d^2 tx + 4 d^2 y^2 - 8 d^2 ty + 8 d^2 t^2. We can cancel some terms, to get

$$16t^2x^2 + 16t^2y^2 + d^4 + 32t^2xy = 4d^2x^2 + 4d^2y^2 + 8d^2t^2.$$

We want this equation to simplify to $xy = 1$. For this to occur, the coefficients of x^2 and y^2 on both sides must be equal, so

$$16t^2 = 4d^2.$$

Then $d^2 = 4t^2$, so $d = 2t$. The equation above becomes

$$16t^4 + 32t^2xy = 32t^4.$$

Then $32t^2xy = 16t^4$, so $xy = \frac{t^2}{2}$. Thus, $t = \sqrt{2}$, so the distance between the foci $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$ is 4.
Note: Students should show all steps in their mathematical solution.

12. **Answer:** 84. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key